

AI Foundations & Developer Environment

Q1. What is Artificial Intelligence?

Artificial Intelligence is a technology that helps computers do tasks which does not need humans to think. AI can make us learn, solve problems, give answers to our questions... Examples of AI are ChatGPT, Google Assistant etc..

Q2. Difference Between AI, Machine Learning, Generative AI, and Agentic AI

AI is the main field that makes machines smart.

Machine Learning (ML) is a part of AI where computers learn from data.

Generative AI creates new content such as text, images, music, and videos.

Agentic AI can plan tasks, make decisions, and perform actions on its own.

Q3. Explain Any Three Open-Source AI Models Available on Hugging Face

Llama 3

Llama 3 is an AI model developed by Meta. It can answer questions, write content, and help with the help coding.

Mistral 7B

Mistral 7B is a fast and lightweight AI model. It is useful for chatbots and text generation.

Stable Diffusion

Stable Diffusion is used to create images from text prompts. Many people use it for AI art and design.

Q4. Difference Between Traditional Programming and AI-Assisted Development

In traditional programming, developers write all the code themselves.

In AI-assisted development, AI tools help developers write code, find errors, and save time.

Traditional programming depends on fixed rules, while AI-assisted development uses intelligent suggestions.

Q5. Explain AI Hallucinations with One Example

AI hallucination happens when an AI gives wrong information but acts as if it is correct.

Example:

If someone asks, "Who was the first person born on Mars?", the AI may give a fake name. This is a hallucination because no one has been born on Mars

Figure 1: AI-generated image showing Artificial Intelligence technology.

